
Chapter 3.

Point 1. Social Media Text Analytics

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Social Media Text Analytics

- Social media text analytics is a technique for **extracting and analyzing business insights** from the textual content of social media platforms.
- It can be used to **understand emotions and sentiments expressed** in user-generated content related to a brand or product.
- Text analytics has roots in various fields such as **data mining** and **natural language processing**.
- It is useful for **quickly and accurately** understanding the **vast volume of social media content generated by users**.
- Approximately **80% of data in organizations is textual**, making text analytics an important tool for business intelligence.

Types of Social Media Text

- Depending on nature of social media text , it can be classified as
 - **Static Text**
 - **Dynamic Text**

Dynamic Text

- **Dynamic text** is **real-time, user-generated text** that expresses about content or information on social media.
- It is typically **shorter in length** and **diverse in nature**, and is often **updated or deleted frequently**.
- Examples of dynamic text include **tweets, Facebook comments, and product reviews**.
- **Dynamic text** is often related to **social, political, economic, personal, cultural, or business issues** and **expresses the views and feelings of the user**.
- **Understanding dynamic text** can provide **valuable insights** into the thoughts and opinions of social media users.



Social Media Dynamic Text Examples:

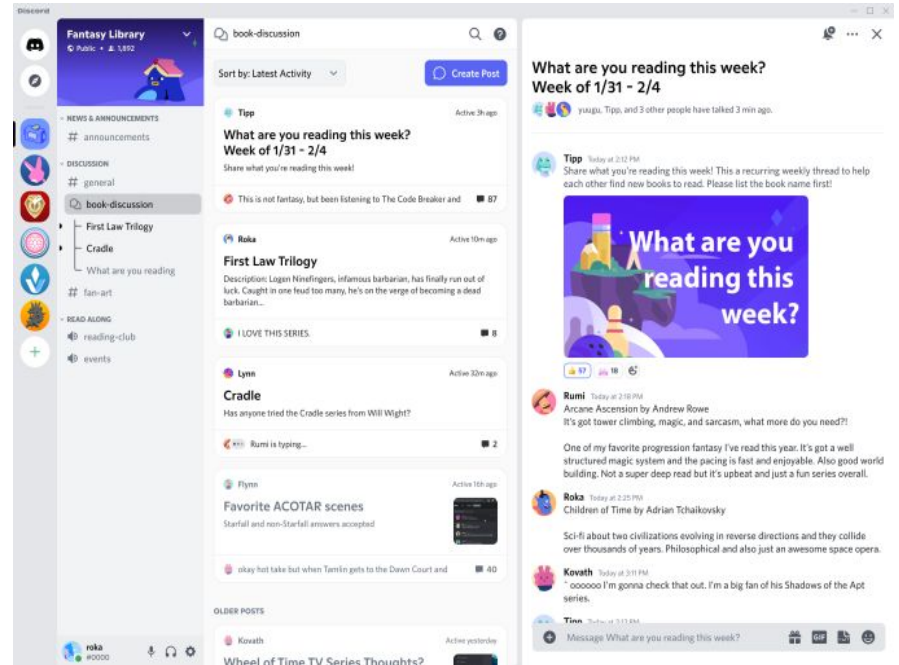
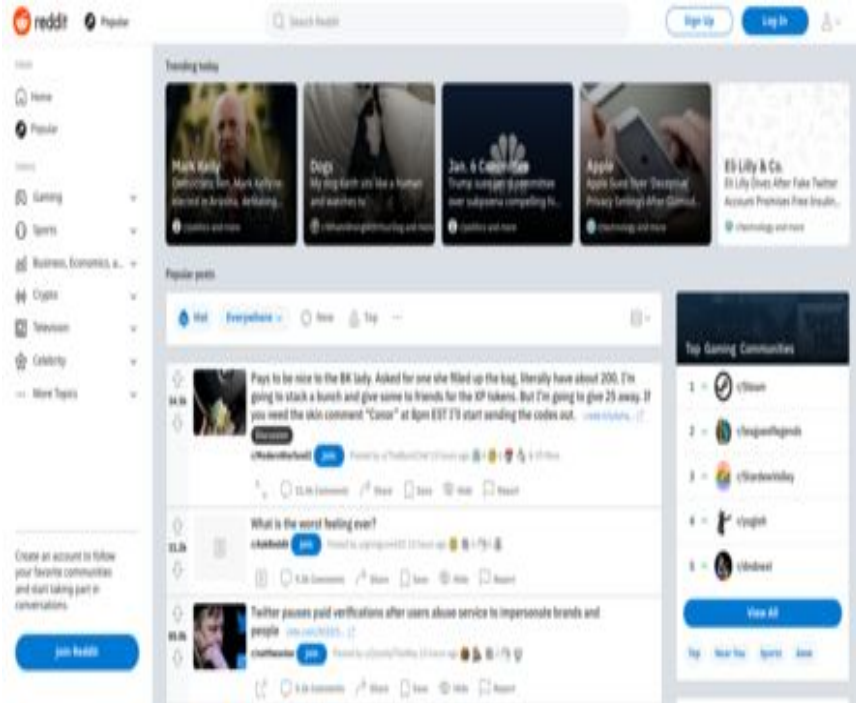
- Social media text includes **tweets, comments, discussions, conversations, and reviews.**
- **Tweets** are short messages (up to 140 characters) posted on Twitter, which may include text, images, video, or links.
- Tweet data provides a **valuable source for mining value business insights, including exploring trending topics, measuring brand sentiment, and gathering feedback on new products and services.**

- **Comments** are written statements expressing opinions about content on social media.
- Like **tweets, social media comments are also a great source for mining valuable business insights from social media** Almost all social media platforms provide commenting features.

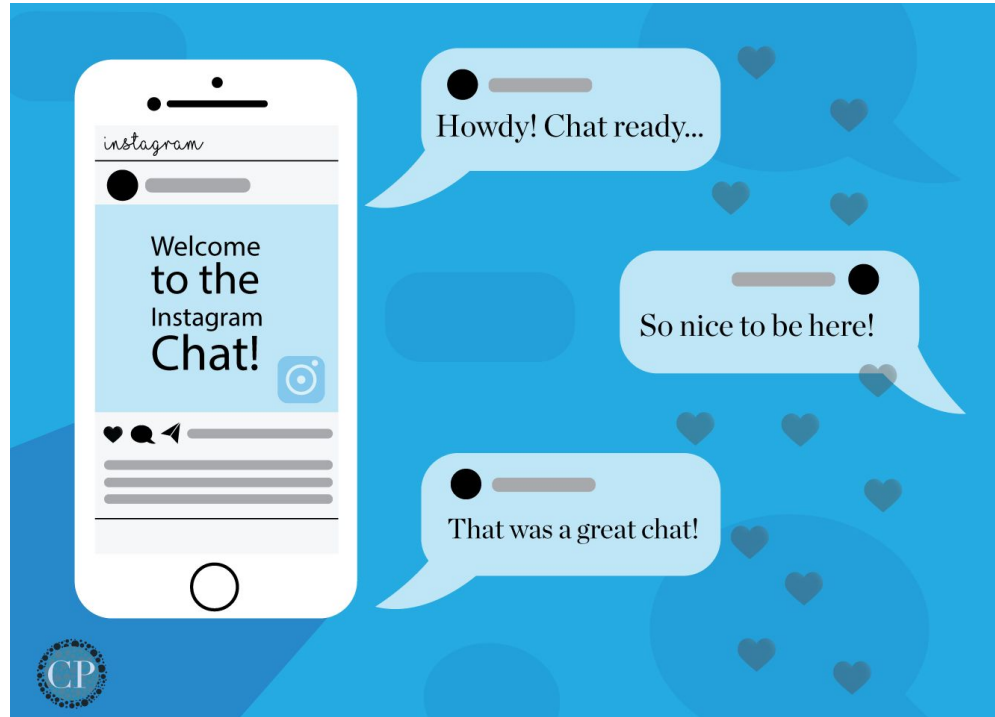


- **Discussions** take place **through internet forums**, which are organized by **topic or thread** and **allow users to post questions and replies**.
- A discussion forum is **an internet-based group**. It can take many forms, including online bulletin boards. **These discussion forums usually allow users to exchange ideas and have conversations**.
- **It's common for websites to add a discussion forum to their website for people to discuss the product, service, or organization and even help each other**.
- **It reduces customer support requests, is great for search engine optimization, and creates a sense of community**.

- Reddit and Discord are examples of Discussion forums
- Quora is an example of a Question-Answer Forum

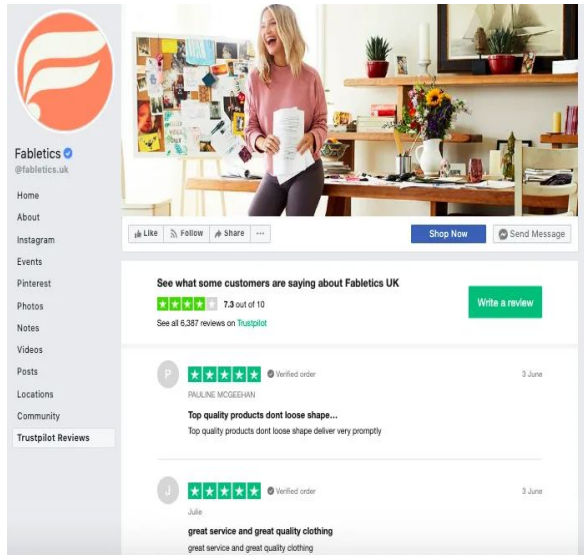


- **Conversations (chatting)** are instant exchanges of written messages between two or more people, often through dedicated messaging tools. Most conversations are private in nature.



- A **variety of messaging tools** have been developed for textual conversation, including **desktop-based (e.g., Skype); web-based (e.g., Google Hangouts and Facebook chat)** and **mobile-based (e.g., Viber, WhatsApp)**.
- **Note that these services are not only limited to textual conversation, but also support video and voice conversation.**
- And now that **all media are converging**, most of the **messaging services also come in desktop, mobile, and web forms**. For example, **Skype has both desktop and Smartphone versions**.

- **Reviews** are **evaluations of products or services by customers or experts, which may include ratings and text.** Product reviews can provide valuable insights into customer opinions and feelings.



Comment vs Review :

- Comment is a few words or sentences with the writers observations on a document or an event.
- A Review is a longer, more formal write-up concerning a position , a theatrical/ cinema performance or someone's work over a set period of time.

Static Text

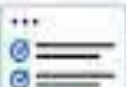
- **Static social media text** is typically **longer in length** and **is generated , updated or deleted less frequently**.
- **Examples** of static text include **wiki content, blog pages, Word documents, corporate reports, emails, and news transcripts**.
- The purpose of static text is often to **inform, educate, and elaborate** on a topic.
- Static text can provide **valuable insights** and information for businesses and organizations.

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December 18, 2018 by [WordPress.com](#)

It's the time of year when we all take a moment to reflect on the past year and look forward to the future. For our friend, the holidays are a time to celebrate the success of Checkidax.com, a business that has grown from a small idea to a thriving enterprise.

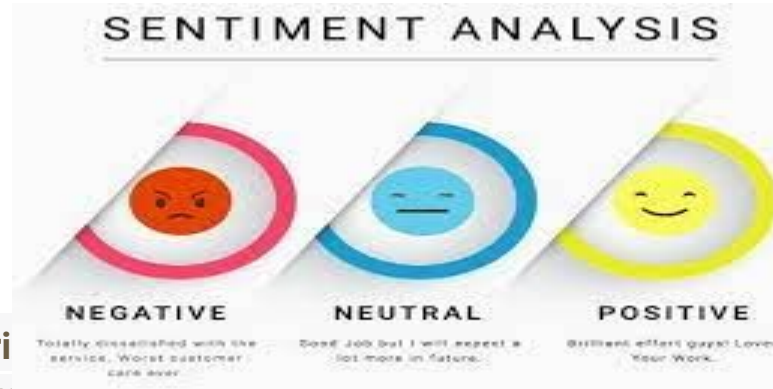
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Purpose of Text Analytics



Figure 4. Purpose of social media text analytics

Sentiment Analysis



- Sentiment analysis involves **category** or neutral. It is often used to **understand how customers feel about a product, service, or issue.**
- Tools like **Semantria** use algorithms to identify sentiment-bearing phrases in text and assign them a score based on a logarithmic scale.
- Scores are then combined to determine the overall sentiment of a document or sentence.
- Sentiment analysis can **provide valuable insights into the emotions and opinions of social media users.**

Sentiment Analysis

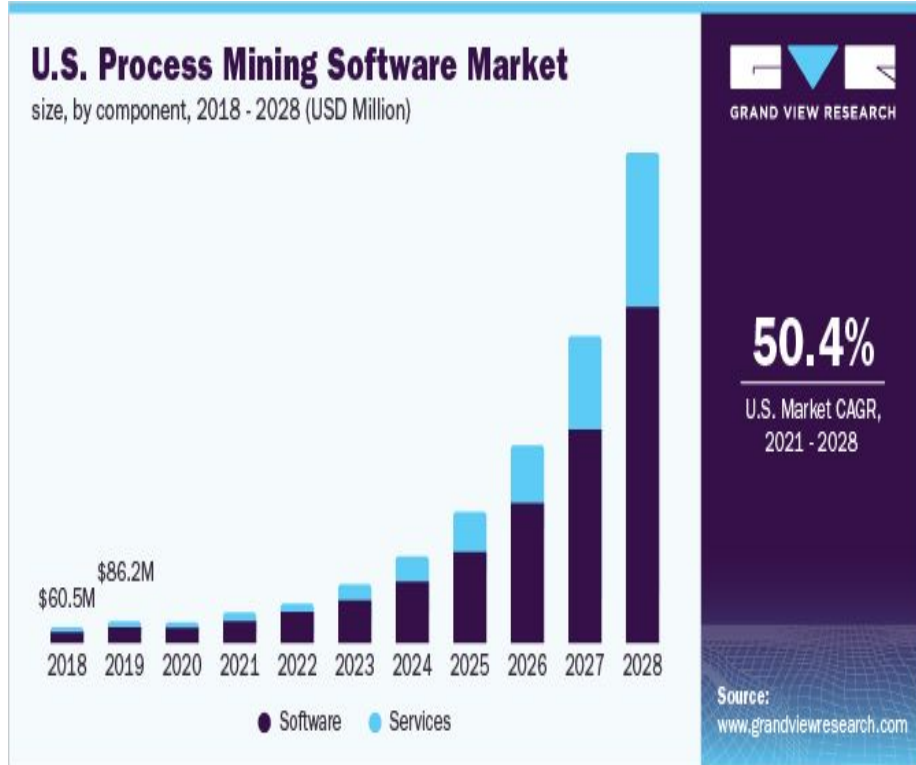
- **Step 1:** It breaks the document into its basic parts of speech, called POS tags, which identify the structural elements of a sentence (e.g. nouns, adjectives, verbs, and adverbs).
- **Step 2:** Algorithms identify sentiment-bearing phrases like “terrible service” or “cool atmosphere.”
- **Step 3:** Each sentiment-bearing phrase earns a score based on a logarithmic scale ranging from negative ten to positive ten.
- **Step 4:** Next, the scores are combined to determine the overall sentiment of the document or sentence. Document scores range between negative two and positive two.

Intention Mining



- Intention mining involves **discovering users' intentions** (to buy) from **natural language social media text** like **users comments, product reviews, tweets and blog posts**.
- Companies **can use intention mining to identify potential customers who intend to buy the products and service existing customers** who have issues with a product.
- Examples of **intention-bearing phrases** include **"buy," "purchase," and "quit."**
- Tools like **Semantria** can be used to mine intentions from social media text.
- **Intention mining** can provide valuable insights into the **needs and desires of social media users**.

Trends Mining :

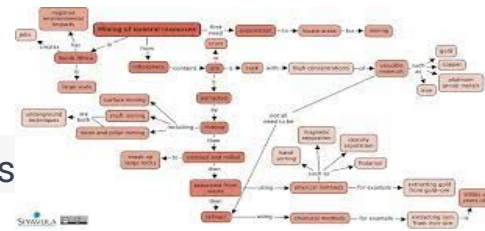


Trends mining, also known as **predictive analytics**, uses large amounts of historical and real-time social media data to predict future events.

- It involves **identifying patterns and trends** in social media data to **improve products, services, or customer satisfaction**.
- Techniques used in trends mining include **machine learning, data mining, and social network analysis**.
- Predictive analytics using business data has been used in various industries, but **social media predictive analytics is still an emerging practice**.
- Trends mining can provide **valuable insights** for businesses and organizations looking to **anticipate the needs and desires of their customers**.

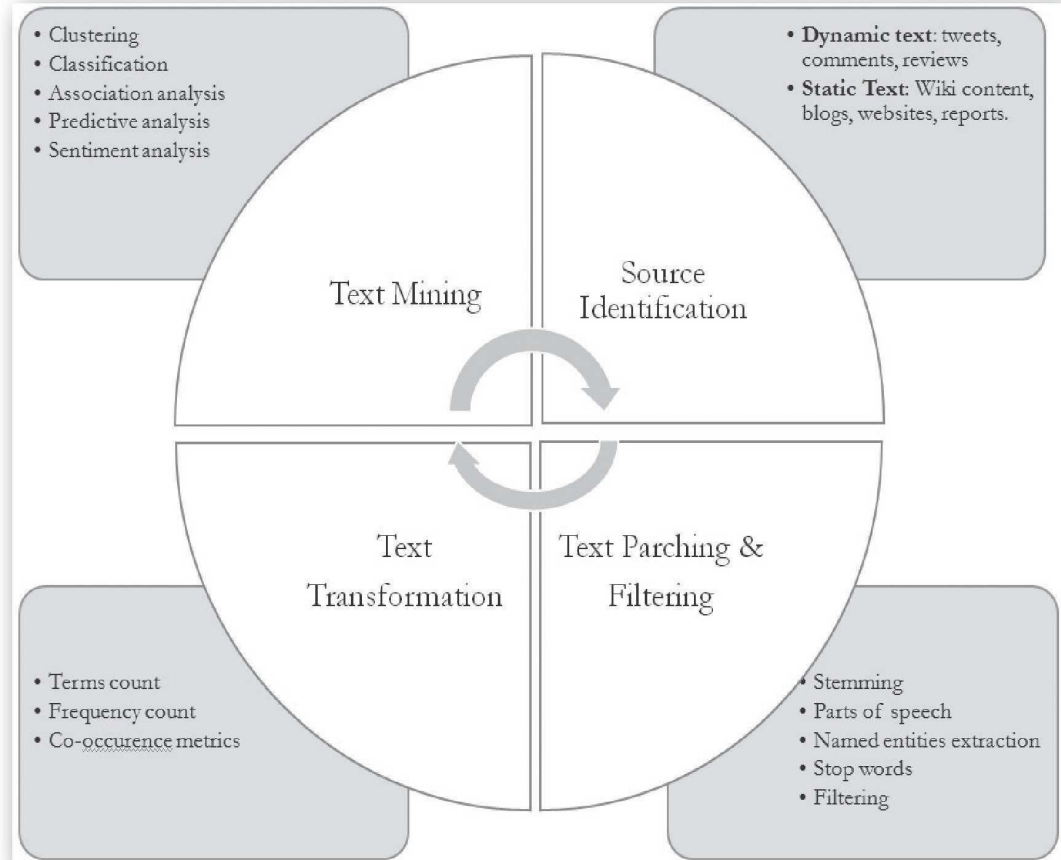
Concept Mining

- Concept mining is a method for **extracting ideas** and **concepts** from documents
- A kind of task derived from the knowledge discovery, focused to recognize patterns in large collections of unstructured data (that is, **linguistic corpora**), which contain relevant information associated to **concepts**.
- It is used to **classify**, **cluster**, and **rank** these ideas.
- Concept mining is different from text mining, which focuses on extracting specific information rather than broader ideas and concepts.
- Concept mining is helpful in a number of applications, including the **evaluation of software programs**, **Detecting and indexing similar documents in large corpora**, **Clustering documents by topic** and in various applications
- Examples of documents that can be analyzed using concept mining include social media text, web pages and news transcripts



Steps in Text Analytics

- Text analytics, like any other form of business intelligence from text, is a process that involves the use of statistical and machine learning techniques to extract meaningful information from text data.
- While the steps required for text analytics are not always the same, a typical social media text analytics process might involve the following steps:



IDENTIFICATION AND SEARCHING :

- The text analytics process starts **with identifying the source of the text that will be analyzed.**
- Text posted on social media is **dynamic, huge, diverse, multilingual, and noisy.**
- Thus, finding **the right source for the purpose of text analytics is very crucial for gaining useful business insights.**
- The **genre of the source text also will determine the type of tool used to extract and analyze it.**
- For example, **extracting tweets requires different tools and approaches than analyzing a document or website text**

TEXT PARSING AND FILTERING :

- The next step is to **parse, clean, and filter the text**, and create a dictionary of words using NPL, which is mostly based on machine learning techniques.
- In order for computer and algorithms to **extract meanings from the text**, the **sentence structures and parts of speech** are determined, **named entities** extracted (people, organizations, product/service names, etc.), **stop words** removed, and spellings are checked.
- Most of these steps are **automatic**; however, in certain stages, human intervention is required.
- For example, in the **filtering stage**, manually cleaning (by humans with domain expertise) may be required to remove unwanted or irrelevant terms.

TEXT TRANSFORMATION :

- For analytical algorithms to be applied to the text, it should be transformed into a computer-readable format (e.g., 0s and 1s) for analysis.
- The cleaned text is thus transformed into numerical representations using linear algebra-based techniques, such as latent semantic analysis and vector space models.

TEXT MINING :

- At this step, **the text is actually mined to extract the needed business insights.**
- Varietiy of text mining algorithms are applied to the text, **such as clustering, association, classification, and predictive analysis, and sentiment analysis.**
- Text analysis employs these sophisticated algorithms to **extract sentiment and meanings from the text in a similar manner to the way human do; however, the process is thousand times faster.**

Text Analytics Tools

- **Discovertext:** Discovertext (<http://discovertext.com/>) is a powerful platform for collecting, cleaning, and analyzing text and social media data streams.
- **Lexalytics:** Lexalytics (<http://www.lexalytics.com/>) is a social media text and semantic analysis tool for social media platforms, including Twitter, Facebook, blogs, etc.
- **Tweet Archivist:** Tweet Archivist (<https://www.tweetarchivist.com/>) is focused on searching, archiving, analyzing, and visualizing tweets based on a search term or hashtag (#).
- **Twitonomy:** Twitonomy (<https://www.twitonomy.com/>) is a Twitter analytics tool for getting detailed and visual analytics on tweets, retweets, replies, mentions, hashtags, followers, etc.
- **Netlytic:** Netlytic (<https://netlytic.org>) is a cloud-based text and social network analytics platform for social media text that discovers social networks from online conversations on social media sites.
- **LIWC:** Linguistic Inquiry and Word Count (LIWC) is a text analysis tool for analyzing emotional, cognitive, structural, and process components present in individuals' verbal and written speech samples: <http://www.liwc.net/>
- **Voyant:** Voyant (<http://voyant-tools.org/>) is a web-based text reading and analysis. With Voyant, a body of text can be read from a file or directly exported from a website.

Case Study

- FlyerTalk.com, is a company that specializes in products and services for frequent travelers. The website features chat boards and discussions about the latest travel information and loyalty programs for airlines and hotels.
- FlyerTalk.com wanted to analyze the large amount of customer feedback and opinions expressed on their website.
- They used the SPSS software for analysis
 - **Data collection** involved using web scraping techniques to crawl and scrape the website for specific information, and store and screen scrape the data into a structured format.
 - **Text coding and categorization** involved assigning codes and categories to the data to make it easier to analyze.
 - Finally, **text mining and visualization** involved using statistical analysis and visual representation of the data to draw insights and conclusions.